1 Solution — GIAM 3.3 Problem 6

1.1 Problem

Prove (by contraposition) that for all integers x and y, if $x+y$ is odd, then $x \neq y$.

1.2 Answer

$$\boxed{\text{Contrapositive: } x = y \implies x+y = 2x, \text{ which is even.}}$$

1.3 Proof

Theorem. *For all integers x and y, if $x+y$ is odd then $x \neq y$.*

Proof. *The statement is logically equivalent to its contrapositive,*

$$\forall x, y \in \mathbb{Z}, \quad x = y \implies x+y \text{ is even,}$$

so we prove that instead. (The negation of $x \neq y$ is $x = y$, and the negation of "odd" is "even", since every integer is exactly one of the two.)

Let x and y be particular but arbitrarily chosen integers, and suppose $x = y$. Substituting $y = x$ into the sum,

$$x + y = x + x = 2x.$$

Since x is an integer, $2x$ is of the form $2k$ with $k = x$ an integer, so $x+y$ is even.

Since x and y were arbitrary, the contrapositive holds universally, and therefore so does the original statement. ■

1.4 The Picture

TO PROVE for all integers x and y : x + y is odd ⇒ x ≠ y

CONTRAPOSITIVE **x = y ⇒ x + y is even**

a "≠" conclusion is hard to attack head-on; flipping turns it into an equation you can substitute with

THE PROOF, IN ONE SUBSTITUTION

x = y ⇒ x + y = x + x = 2x ⇒ x + y is even ■

the only fact used is that 2x is an even integer whenever x is an integer

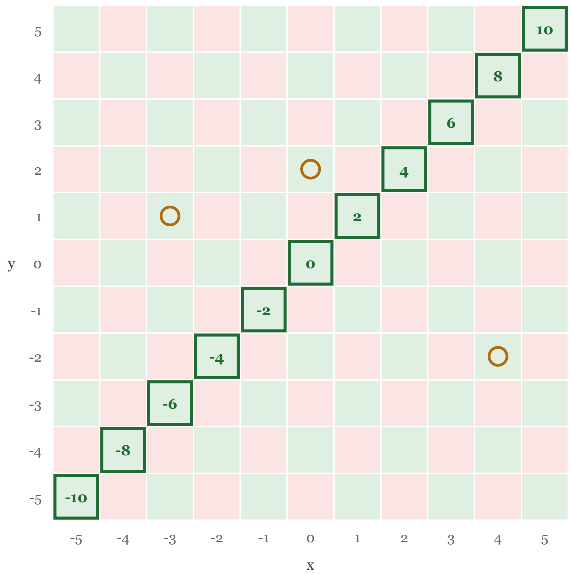
THE PARITY CHECKERBOARD

Each cell is a pair (x, y).
Green: x + y even.
Pink: x + y odd.

The theorem says no pink cell sits on the diagonal x = y.

The outlined diagonal cells carry their sum 2x — all even, so all green.

Circled: pairs with x ≠ y but x + y even. The converse is false.



THE CONCLUSION IS MUCH WEAKER THAN WHAT IS TRUE

x + y odd ⇒ x − y odd ⇒ x and y have opposite parity ⇒ x ≠ y

only the last arrow is one-way, and it is the one the problem asks for — the theorem throws information away

It breaks for three summands:
1 + 1 + 1 = 3 is odd, yet all three are equal. For n copies of x the sum is nx, which can be odd only when n is odd — so "odd sum forces them not all equal" is a theorem exactly for even n. Checked for n up to 7.

Figure 1.1: A diagram in several bands. The first states what is to be proved — for all integers x and y, x plus y odd implies x is not equal to y — and its contrapositive, x equals y implies x plus y is even, noting that a not-equal conclusion is hard to attack head-on and flipping turns it into an equation you can substitute with. The second band gives the proof in one substitution: x equals y implies x plus y equals x plus x equals 2x, which is even; the only fact used is that 2x is an even integer whenever x is an integer. The main image is a parity checkerboard: an eleven-by-eleven grid of cells, one for each pair of integers x and y from minus five to five, coloured green when x plus y is even and pink when x plus y is odd, producing a checkerboard pattern. The cells along the diagonal where x equals y are outlined in green and labelled with their sums, minus ten, minus eight, minus six, minus four, minus two, zero, two, four, six, eight, ten — all even, hence all green. Three cells off the diagonal are circled in amber to show pairs with x not equal to y whose sum

is nevertheless even, demonstrating that the converse is false. A band below shows the chain: x plus y odd implies x minus y odd implies x and y have opposite parity implies x is not equal to y; only the last arrow is one-way, and it is the one the problem asks for, so the theorem throws information away. A final band notes that the result breaks for three summands: 1 plus 1 plus 1 equals 3 is odd yet all three are equal, and for n copies of x the sum is nx, which can be odd only when n is odd, so the statement is a theorem exactly for even n, checked for n up to 7.

1.5 Remark — Why Contraposition Is the Right Tool

The conclusion of the original statement is $x \neq y$ — a **negation**. That is what makes a direct attack awkward: from "$x + y$ is odd" you would have to argue that x and y *fail* to be equal, and non-equality gives you nothing to compute with.

Contraposition swaps the two negations for their positive forms and, crucially, promotes the conclusion into a hypothesis:

$$\underbrace{x + y \text{ odd} \implies x \neq y}_{\text{conclusion is a non-fact}} \qquad \text{becomes} \qquad \underbrace{x = y \implies x + y \text{ even}}_{\text{hypothesis is an equation}}.$$

Now $x = y$ is an *equation*, and an equation licenses substitution. One substitution later the proof is finished. This is the same lesson as Problems 1 and 2 of this section, in its purest form: the technique is not really "flip the implication", it is "arrange for the hypothesis to be something you can do algebra with".

Note also that the flip relies on the parity dichotomy — every integer is odd or even and never both — to turn "not odd" into "even". As in Problem 1, that is a separate fact about $\mathbb{Z}$ and not part of contraposition itself.

1.6 Remark — The Converse Is False

The theorem is a one-way implication, and the checkerboard makes this vivid. Consider

$$x = 1, \quad y = 3 \qquad \implies \qquad x \neq y \quad \text{but} \quad x + y = 4 \text{ is even.}$$

Within the small grid $0 \leq x, y \leq 7$ there are 24 such pairs. So $x \neq y$ does **not** imply $x + y$ is odd, and the biconditional fails.

This is worth stating plainly because the contrapositive of a statement is equivalent to it, while the *converse* is a different statement entirely — a distinction easy to blur when both involve rearranging the same two clauses. Here:

	statement	**true?**
original	$x + y$ odd $\implies x \neq y$	yes
contrapositive	$x = y \implies x + y$ even	yes — it is the same statement
converse	$x \neq y \implies x + y$ odd	**no**
inverse	$x + y$ even $\implies x = y$	**no — it is the converse flipped**

1.7 Remark — The Theorem Discards Most of What Is True

The hypothesis $x + y$ odd is far more informative than the conclusion $x \neq y$ admits. Since $x + y$ and $x - y$ differ by $2y$, they always have the same parity, so

$$x + y \text{ odd} \iff x - y \text{ odd} \iff x \not\equiv y \pmod 2 \iff \text{exactly one of } x, y \text{ is odd},$$

all verified exhaustively over $-60 \leq x, y \leq 60$. The full chain of consequences is

$$x + y \text{ odd} \implies x - y \text{ odd} \implies x \not\equiv y \pmod 2 \implies x \neq y,$$

and every arrow but the last is reversible. The problem asks for the weakest link in the chain — the one place information is genuinely lost, since $x \neq y$ permits $x = 1, y = 3$ while $x \not\equiv y \pmod 2$ does not.

Seen this way, "$x \neq y$" is a shadow of the real content: x and y sit in *different parity classes*, which is a much stronger separation than merely being different numbers. The checkerboard shows it — the pink cells are not just off the diagonal, they avoid every diagonal of the same colour.

1.8 Remark — It Fails for Three Summands

A natural generalisation would be: *if a sum of integers is odd, they are not all equal.* This is **false** as soon as there are three of them:

$$1 + 1 + 1 = 3 \quad \text{is odd, yet } x = y = z = 1.$$

The reason is transparent once the proof is written out. With n copies of x the sum is nx, and

$$nx \text{ is odd} \iff n \text{ is odd and } x \text{ is odd.}$$

So the statement "odd sum $\Rightarrow$ not all equal" is a theorem **exactly when n is even** — confirmed for $n \leq 7$. The original problem is the case $n = 2$, and in the line $x + y = 2x$ the coefficient 2 is doing more work than it appears to: it is not merely arithmetic, it is the reason the sum is forced even.

That is a good instinct to develop when reading a short proof. The line $x + y = 2x$ looks like trivial arithmetic, but the specific number 2 appearing there is precisely what makes the parity argument close, and a version of the problem with three variables would need a different theorem.